

vapor

A certified tensor compiler and the laboratory built on it

Architecture, semantics and guarantees

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Technical monograph — version 0.17

vapor is a compiler and runtime for tensor programs, written in Elixir on the BEAM with a small native worker in Zig, and a laboratory of tools built on top of it. Its organising principle is that every result carries what is needed to judge it. Tensor programs compile to machine code for x86-64 (AVX2, AVX-512), AArch64, RISC-V with the vector extension and Vulkan, and under the canonical policy produce the same bits on all of them and on an exact binary32 oracle. Register allocation is checked by a verifier proved correct in Lean 4 and extracted to Elixir. A six-rung verification ladder ends in an Ed25519 certificate that independent nodes can reproduce byte for byte and co-sign. Models enter through an airlock that separates the function a network computes from the way a family spells it; network access enters through a second airlock that only the person can open. Around the core sit deciders whose answers come with independently checked certificates (exact linear and integer programming, SAT with DRUP proofs, Gröbner bases, causal identification, polynomial positivity, circuit equivalence), a bilingual language of decided claims, a search engine whose results are re-verified, and an evidence discipline in which every quality check is paired with a control. This monograph describes the system as it is, its architecture, the invariants each component keeps, how they are checked, and the limits that remain.

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1 Introduction

1.1 What the system is

vapor has two parts that share one discipline.

The first is a **compiler and runtime for tensor programs**. A program is a term of a small free algebra (element-wise operations, contractions, reductions, the operators of modern neural networks) whose meaning is fixed by an exact reference implementation, the *oracle*. The compiler lowers a program to a portable kernel IR, allocates registers without spilling, and encodes machine code for four instruction sets and for Vulkan compute shaders. The runtime executes that code in a separate, sandboxed process, keeps model weights in content-addressed shared memory, serves models through an OpenAI-compatible interface and records a receipt for every answer.

The second is a **laboratory**: deciders, languages, search, numerical workbenches, statistical tools, document readers and desks for engineering and finance, all reachable from one command-line interpreter, a browser console, an MCP server for agents, a language server for editors, and a small editor of its own.

What the two share is the rule that a result is never just a value. A compiled kernel comes with the evidence that it computes the oracle’s bits; an *optimal* integer program comes with a branch-and-bound tree that separate code checks; a refuted conservation law comes with the point that refutes it; a statistical claim comes with a control that a broken implementation would have passed.

1.2 Three properties

Three properties run through every chapter.

Determinism. Under the *canonical* numerical policy, a program produces the same bits on every substrate: x86-64 with AVX2 or AVX-512, AArch64 with NEON, RISC-V with RVV (native or interpreted), Vulkan (on any driver) and the oracle. The bits also do not depend on the number of threads, on how many sequences share a decoding step, on where the key-value cache lives, or on which replica serves a request. Parallelism is drawn from independent outputs, never from reassociating a sum (Chapter 3).

Certification. Wherever a claim can be checked by a procedure simpler than the one that produced it, the system ships the certificate and the checker. Register allocations are revalidated by a checker proved sound in Lean 4. Kernels climb a ladder of differential checks that ends in a signed, reproducible certificate. Deciders return witnesses, Farkas vectors, DRUP proofs or hedges, and each is checked by code that shares nothing with the search (Chapters 6 and 9).

Containment. Nothing generated runs inside the BEAM: there are no NIFs. Machine code runs in a static, libc-free worker under seccomp, with W^X pages and a watchdog. Work on untrusted input runs under one containment primitive, the *hermetic seal*, which caps memory (on-heap and off-heap) and time. Models enter only through the model airlock; bytes from the network enter only through the network airlock, and only a person can open it (Chapters 7 and 8).

1.3 How to read this document

Chapter 2 gives the layers and the trust boundaries. Chapters 3 to 6 describe the core: the algebra and its numerical semantics, the compiler, the runtime and the verification ladder. Chapters 7 and 8 describe the two airlocks. Chapters 9 and 10 describe the certified deciders and the languages built on them. Chapter 11 describes how quality is measured and how the test suite is organised, Chapter 12 the interfaces, and Chapter 13 what the system does not do. The appendix maps components to modules.

Every statement of fact in this document corresponds to a test or a check in the repository, named in the system's own documents (`docs/`); this monograph gives the architecture and leaves the contestation commands to them.

2 System overview

2.1 Layers

Figure 2.1 shows the layers. Each layer uses the ones below it and adds no trust to them: what a layer produces is either a certified program for the core or canonical data.

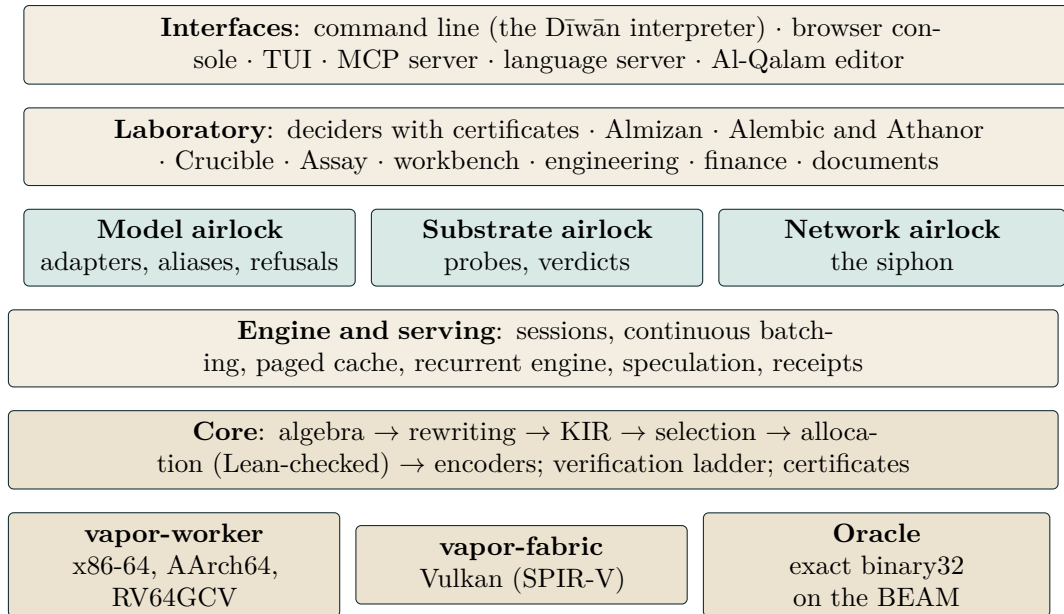


Figure 2.1: Layers. The airlocks are the only ways in for models, substrates and network bytes.

The core turns a term of the algebra into machine code and evidence. It is about ten thousand lines of Elixir (algebra, compiler, kernel IR, encoders, verification, runtime), plus the code extracted from Lean and the native worker.

The engine runs compiled model programs in resident sessions: one compilation, many steps, weights mapped once. It schedules many sequences into one step without changing any sequence’s bits.

The airlocks are the boundaries through which outside things enter. A checkpoint must be claimed by an adapter that writes its function in the algebra; a substrate must pass numerical probes before it receives canonical programs; a byte from the network must come through a fetcher that the person declared and ran.

The laboratory holds tools whose answers carry their evidence. Some compile to the core (a Monte Carlo pricer whose generator is written as algebra terms, so the same bits come out on every substrate); most inherit only its canonical encodings, Merkle trees and the discipline of independent checking.

2.2 Trust boundaries

boundary	what crosses it, and what keeps it
BEAM $\leftrightarrow$ native code	framed messages on stdio (<code>{packet, 4}</code>); no NIF is loaded (a test forbids it); a crash, an illegal instruction or a timeout kills the worker process, which is restarted, and the next unit runs
worker $\leftrightarrow$ kernel	seccomp-BPF allow-list, <code>PR_SET_NO_NEW_PRIVS</code> , <code>W^X</code> code pages, a <code>SIGALRM</code> watchdog
job $\leftrightarrow$ host	the hermetic seal: a process with a heap cap that counts off-heap binaries, and a deadline
checkpoint $\leftrightarrow$ engine	the model airlock: claimed by an adapter, every tensor accounted for, configuration keys that change the mathematics read or refused by name
substrate $\leftrightarrow$ dispatcher	the substrate airlock: probes classify a substrate as canonical, within an envelope, or refused
network $\leftrightarrow$ system	the siphon: only fetchers the person declares, run by the person; an agent can only queue a request
agent $\leftrightarrow$ world	effect classes (<code>pure</code> , <code>observe</code> , <code>act</code>), grants fixed in the agent’s digest, a write-ahead journal

Table 2.1: Trust boundaries and what holds each.

The trusted base is the BEAM, the Linux kernel, the IEEE-754 standard model of rounding (from which the error bounds are proved), the printer of the Lean extractor (about two hundred lines), and the Vulkan driver, which runs in its own process.

2.3 Size

component	lines
Elixir, all of <code>lib/</code> (379 files, 425 modules)	101,000
of which the core (algebra, compiler, KIR, encoders, verification, runtime)	10,400
native worker and Vulkan daemon (Zig, <code>native/src</code>)	4,700
Lean 4 proofs and extractor (82 theorems, no <code>sorry</code> , no <code>axiom</code>)	1,500
tests (Elixir, and the Python and Node scripts of the differential tiers; 159 files)	27,000

Table 2.2: Size of the system. The native binaries are 330–360 KB each; the BEAM application has no dependencies outside OTP and Elixir.

A full build takes about thirty seconds on two virtual CPUs. The application declares no Hex dependency: JSON, CBOR, SHA-256 trees, Ed25519, safetensors, GGUF, PNG, JPEG, PDF and the HTTP server are implemented in the tree or taken from OTP.

3 The algebra and its numerical semantics

3.1 Programs are terms

A program is a term of `Vapor.Algebra.Term`, a *symbolic* free algebra: every operator is a name whose meaning is fixed by the oracle (`Vapor.Runtime.Oracle`), never an embedded closure. That is what makes a term compilable to machine code and comparable across substrates.

- **Generators:** inputs and constants; element-wise operations (`add`, `sub`, `mul`, `fma`, `neg`, `relu`, the canonical transcendental functions, the comparison-select `sel(a,b,x,y) = a < b ? x : y`); contractions (`linear(x,w) = xwT`, grouped and masked variants, 4-bit and int8 matrices); reductions that keep rank; and the operators of neural networks (`gather_row`, `rope`, `kv_write` and `attention` with an explicit scale, contiguous and paged, `sample`).
- **Semi-dynamic dimensions** `{:dyn, :S, max}`: every bound is established at the maximum, and monotonicity lemmas proved in Lean carry the certificate to every extent $S \leq \text{max}$.
- **Recurrent programs** declare state feedback (`state: [h: :h_next]`): the whole loop runs in the worker behind one crossing from the BEAM, the state never leaves the worker, and outputs stream per step.

Numerical data live in contiguous binaries (`Vapor.Tensor`); random access uses tuples.

3.2 Exact binary32 on the BEAM

`Vapor.F32` implements binary32 exactly: values are bit patterns; `add`, `sub` and `mul` go through binary64 and one rounding (correct, since $53 \geq 2 \cdot 24 + 2$); `fma` is computed in dyadic rationals with a single rounding, round-to-nearest-even, gradual underflow and overflow to infinity. It agrees with the hardware on 800,000 operations including subnormals and signed zeros. The oracle is built on it.

3.3 Two policies

policy	semantics	guarantee
<code>:canonical</code>	multiply and add rounded separately; every reduction in one fixed 16-lane tree ($r_8 \rightarrow r_4 \rightarrow r_2 \rightarrow r$); division correctly rounded; transcendental functions as fixed micro-programs	identical bits on x86-64, AArch64, RVV, the interpreter, Vulkan and the oracle
<code>:fast</code>	fused multiply-add where the operator allows	every output inside a rigorous error envelope on every substrate

Table 3.1: Numerical policies.

The central idea is that reproducibility does not require slowness when parallelism comes from independent outputs. Each row of a matrix–vector product has exactly sixteen accumulators reduced in the fixed tree, whatever the vector width; throughput comes from processing several rows per iteration (chosen by the register allocator per instruction set), which changes no bit.

Under the canonical policy $+$, $-$, $\times$ and $\div$ are IEEE operations on every substrate (with the GPU convention of flushing subnormals for division: DAZ/FTZ). Division is computed by Markstein’s method with Dekker’s exact residual and no fused operation; the logarithm is within one ulp of the correctly rounded value; the other transcendental functions are within one or two ulps, identical everywhere. Tables that must be exact, such as rotary position frequencies, are computed by a correctly rounded routine (**Vapor.CR**, Ziv’s strategy).

Semantics version. The bits of a program are a function of its terms *and* of the definition of the canonical functions. Certificates record **Vapor.Canon.version/0**, so a change in a canonical function cannot silently change what a certificate means.

3.4 The error envelope

For the fast policy and for substrates that are not canonical, **Vapor.Verify.Envelope** computes, for every output, the exact real value and a rigorous bound valid for *any* conforming substrate (correct rounding, fused or unfused multiply–add, any reduction order, gradual underflow or flush-to-zero), in exact dyadic rationals. Contractions over exact inputs take the Wilkinson form $\gamma_n S + a$, decided by **within_envelope/4** (extracted from Lean); other operations use running error analysis. The verifier never rounds; bounds are only ever shortened upward. The Higham form of γ_n , a bound on products of $(1 + \delta_i)$, is proved in Lean from the standard model $\text{fl}(x \text{ op } y) = (x \text{ op } y)(1 + \delta)$, $|\delta| \leq 2^{-24}$.

3.5 Rewriting without error

Vapor.Compile.Rewrite admits only identities that hold bit for bit over all of IEEE-754, signed zeros included: $\text{neg}(\text{neg } x) \rightarrow x$, $x \cdot 1 \rightarrow x$, $x + (-0) \rightarrow x$, $x - (+0) \rightarrow x$, but not $x + (+0) \rightarrow x$, because $(-0) + (+0) = +0$. Constant folding evaluates the declared semantics themselves. Rewriting therefore has error $\varepsilon = 0$ by construction.

4 The compiler

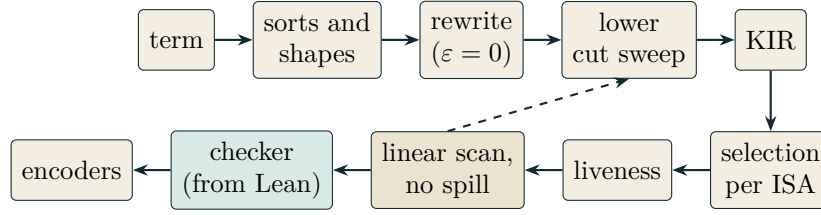


Figure 4.1: The compilation pipeline. An allocation that runs out of registers is never spilled: the compiler goes back (dashed) and retries with a smaller group factor, fewer rows per iteration, or a cut of the fused region.

4.1 A portable kernel IR

Kernels (`Vapor.KIR.Kernels`: fused element-wise strips, reductions, matrix–vector products in f32, bf16 and 4-bit formats, int8 matrix products, and the model operators) are written once over virtual registers with *types*: `:strip`, `:f16l` (sixteen f32 lanes), `:i32acc` and others. Each backend sizes the types. The **group factor** g generalises the RVV register-group multiplier (LMUL) to every instruction set: an AVX2 strip with $g = 4$ is four `ymm` registers in lockstep, an RVV strip with $g = 8$ is an `m8` group.

type	RVV	AVX2	AVX-512	NEON
<code>:strip</code>	mg	g <code>ymm</code>	g <code>zmm</code>	g <code>q</code>
<code>:f16l</code>	<code>m4</code> , <code>vl=16</code>	2 <code>ymm</code>	1 <code>zmm</code>	4 <code>q</code>
<code>:i32acc</code>	$m4g$	g <code>ymm</code>	g <code>zmm</code>	2 <code>q</code>

Table 4.1: One kernel IR, four register files.

Selection happens *before* allocation, as in production compilers. Lane-wise instructions are emitted as bundles (one allocation unit), so a destination can reuse a source register that dies at that point. On AVX-512, strip tails are one masked pass (`bzhi` into mask registers, masked loads and stores only) instead of a scalar loop, and four-byte constants use embedded broadcast; the reduction tree is lane for lane the same as AVX2’s, so the bits are the same.

4.2 Allocation without spill, checked

`Vapor.KIR.Liveness` computes liveness by fixed point over the control-flow graph (values live on a back edge cover the whole loop), with early-clobber constraints for the overlap rules of RVV’s widening instructions. `Vapor.KIR.RegAlloc` is a linear scan with aligned blocks of one, two, four or eight registers.

There is no spill path. When registers run out, the allocator returns the pressure point, and the compiler tries a smaller group factor, then fewer rows per iteration for the 4-bit products, then a cut of the fused region at that point. Every accepted allocation is revalidated by `check_alloc/3`, which is extracted from Lean together with its soundness theorem (`checkAlloc_sound`): groups

are aligned, inside the register file, outside the reserved registers, and no two simultaneously live values share a register. Every emitted region is therefore free of spill and clobber by construction, not by trust in the heuristic.

4.3 Encoders

target	encoding
x86-64 (AVX2, AVX-512)	REX, three-byte VEX, EVEX (always <code>disp32</code>), ModR/M, SIB; System V ABI, callee-saved registers saved only when used
AArch64 (NEON)	A64 words; AAPCS64, <code>x18</code> never allocated
RV64GCV	<code>vtype</code> fields, vector loads and stores with width fields, static round-to-nearest for scalar FP; full psABI; long branches as an inverted branch over a <code>jal</code>
SPIR-V (Vulkan)	a symbolic assembler that interns types and constants and orders the logical sections; <code>NoContraction</code> on every multiply and add under the canonical policy
MSL, StableHLO	emitted for Metal and for PJRT devices, and admitted only through the substrate airlock

Table 4.2: Encoders. The product never calls an assembler.

The tests validate every emitted instruction (every kernel constructor, in every backend, at every group factor and policy) against GNU binutils, and every SPIR-V module against `spirv-val`. Those tools are used only to check; the product encodes bytes itself.

4.4 Cost model

The arbiter predicts the time of a schedule on a substrate as

$$t = \sum_{\text{loops}} \max\left(\frac{F}{\pi}, \frac{Q}{\beta}, \frac{I}{\iota}\right) + \text{overheads},$$

with floating-point work F and bytes Q counted from the schedule and I the static count of instructions in each hot loop as emitted, times its iterations. The third ceiling makes the model honest: the 4-bit matrix–vector product is bound by instruction issue (about 1.1 instructions per weight), not by bandwidth, and a two-ceiling model misses it by a factor of about five. Profiles are declared per vendor; the dispatch decision is the argmin of the predicted time, taken on each node from the certified work and the node’s own profile.

5 Substrates and the runtime

5.1 The worker

`vapor-worker` is a static Zig executable with no `libc`. At start-up (`HELLO`) it fixes the number of threads; the thread pool and the event counters (`perf_event_open`: cycles, instructions and cache events when a PMU exists; task clock, page faults and context switches always) are created *before* the seccomp filter, which is installed with `TSYNC` on every thread. Each call carries a *partition descriptor* (a count argument, a grain, pointers that advance per unit, per-thread scratch, an optional guard) and the worker splits the units among threads. Because the units are independent rows, the result is the same for one thread or many.

Code runs in one of two modes. **Native**: the machine code is copied to a writable page, the page is made read-execute (`W^X`), the instruction cache is synchronised where the architecture needs it, and the kernel is called as `void k(const uint64_t *args)`. **Emulated**: RV64GCV code is interpreted by `rvemu.zig`, which checks every memory access against the bound buffers and every fetch against the code, and turns an unknown instruction into an error with its address and word. The interpreter implements RVV 1.0 faithfully (round-to-nearest, fused `vfmacc`, canonical NaN, NaN-boxing, group alignment, `vll`), with a *poison* mode that writes ones into tail- and mask-agnostic elements, proving that kernels do not depend on them, and a configurable `VLEN` from 128 to 512 bits.

Containment is a seccomp-BPF allow-list (`read`, `write`, `openat`, `close`, `lseek`, `mmap`, `munmap`, `mprotect`, `clock_gettime`, `setitimer`, `signals`, `exit`; the architecture is audited), `PR_SET_NO_NEW_PRIVS`, and a `SIGALRM` watchdog for native code that does not finish. The tests provoke all four failure classes (`SIGILL`, `SIGSEGV`, `SIGALRM`, `SIGSYS`); in each, only the worker dies, the BEAM process that owns it reports the signal, restarts it, and the next unit runs.

5.2 The Vulkan daemon

`vapor-fabric` drives Vulkan compute through hand-written bindings (`libc` only for the loader's `dlopen`): instance, physical device with a compute queue, logical device, buffers, shader modules from the SPIR-V emitted by the BEAM, pipelines with push constants, descriptor sets, one command buffer per run with barriers, per-iteration windows and state copies, submission with a fenced deadline, and readback. A driver crash or a lost device kills only the daemon. Its plans are self-contained, so a restarted daemon needs no replay.

5.3 One protocol

Both processes speak the same framing: `{packet, 4}` frames on standard I/O (big-endian length, little-endian fields), so a process that dies in the middle of a frame cannot desynchronise the BEAM.

frame	code	content
HELLO	1	flags, threads $\rightarrow$ architecture, sandbox state, version, threads
RUN	2	mode, VLEN, flags, fuel, deadline, code, buffers (inline or mapped files), calls with arguments and partition descriptors, iterations, streamed emissions, copies, returns
EMIT	3	per-iteration output, streamed
DONE	4	elapsed time, retired instructions, counters, returned buffers
ERR	5	code, program counter, instruction word, message
OPEN, STEP, CLOSE	6–8	resident sessions: weights mapped once, state kept between steps, each step writing only its inputs and returning only the requested outputs, with optional state copies inside the worker

Table 5.1: The worker protocol. Tables are sized by the frame itself, so a hostile count cannot allocate more than the frame describes.

5.4 Memory

Weights of 64 KiB or more are written once to `/dev/shm/vapor-<sha256>` (content addressed, atomic write) and mapped copy-on-write by every worker; the Vulkan daemon imports them without copy through `VK_EXT_external_memory_host` when the device offers it. Replicas and restarts map the same files, so n workers serving one model occupy one copy in the page cache. Weights can stay in bfloat16: the kernels widen sixteen weights exactly on load, so the result is bit for bit that of the f32 program over the rounded values. Per-token inputs are windows into one buffer; state feedback ($h \leftarrow h_{\text{next}}$) is a copy declared in the plan.

5.5 Dispatch and failover

`Vapor.Runtime.Dispatch` tries the arbiter’s choice and walks down the chain

fabric $\rightarrow$ host AVX-512 $\rightarrow$ host base ISA $\rightarrow$ RVV interpreter $\rightarrow$ oracle,

recording the reason for each step. Under the canonical policy every link computes the same bits, so rerouting never changes an answer.

5.6 Parallelism without changing bits

Speculative decoding needs no further correctness argument: the $k+1$ -row verification step is bit-identical to $k+1$ single steps by batch invariance, so the output equals the target’s greedy decoding whatever the draft, and the draft only changes the number of steps. Block drafters and tree drafts are degenerate and general cases of the same rule.

5.7 The engine and serving

`Vapor.Engine` serves compiled models in resident sessions with continuous batching, a paged key–value cache, replicas and cancellation: the engine monitors the process that asked for each sequence, and a sequence whose requester dies leaves the batch at the next step boundary

form	where	invariant
SIMD	AVX2, AVX-512, NEON, RVV (VLEN 128–512), SPIR-V	same bits as the oracle
instruction-level	R rows per iteration in matrix–vector products, chosen by the allocator	same
threads	the worker’s pool; decode attention split by key–value head	1 thread = n threads
continuous batching	decode and prefill chunks in one step	a sequence alone = in a batch
paged cache	paged key–value writes and attention over a block table	paged = contiguous
replicas	one compilation, shared weight pages, shortest queue	same answer from any replica
speculation	a draft proposes, the target verifies $k+1$ rows in one step	output = the target’s
GPU	SPIR-V for every model operator	fabric = oracle

Table 5.2: Every form of parallelism draws on independent outputs.

without changing the bits of the others. Recurrent models (Mamba, Mamba-2, the delta-rule hybrid of Chapter 7) run as step programs whose state stays in the worker. The HTTP server speaks the OpenAI chat and completion protocols with streaming, and every response carries a receipt, the canonical digest of model, prompt tokens, parameters and generated tokens. Constrained decoding (**Vapor.Grammar**) restricts generation to a grammar derived from a JSON Schema, a regular expression or a tool’s declared arguments, with the invariant that no accepted prefix is a dead end.

6 Verification and certificates

6.1 The ladder

A program is certified by climbing six rungs. Each rung checks something the previous one cannot.

rung	establishes	how
1	sorts, shapes, feedback	<code>Program.check/1</code>
–	exact rewriting	only identities valid over all of IEEE-754
2	admission	the allocation accepted by the extracted checker on <i>every</i> instruction set; int8 contractions free of wrap-around at the maximum extents (<code>admissible/3</code>); SPIR-V limits
3	adjoint identity	$\langle Wx, v \rangle = \langle x, W^T v \rangle$ between the compiled kernel and the exact transpose: catches layout and transposition errors
4	differential oracle	each CPU substrate equal to the exact oracle, bit for bit
5	parity and envelope	equal FNV-1a digests across substrates; every output inside the envelope
6	proof-carrying code	an Ed25519 certificate with a quorum of co-signatures

Table 6.1: The verification ladder.

6.2 The certificate

The payload holds the SHA-256 of the program, of each machine-code blob and of each SPIR-V module; the digest of the Lean sources of the checkers; the evidence of each rung; and the *counted work* (floating-point operations, bytes, instructions per instruction set). It holds nothing that depends on the host: no timings and no dispatch decision. Independent nodes that redo the ladder therefore produce payloads identical byte for byte and can co-sign them (`Certificate.cosign/3`); `verify/3` requires a quorum of valid signatures from distinct trusted keys. A bundle packages artefacts and certificate, and an edge node checks signatures and hashes in about two milliseconds and runs the code without redoing the ladder.

6.3 What is proved in Lean

The Lean 4 development uses the core library only, with no `axiom` and no `sorry`. It proves, among its 82 theorems:

- in $\mathbb{Z}/2^{32}\mathbb{Z}$: associativity and commutativity, that wrapping is a homomorphism, and that *every* reduction tree with a wrapping adder computes the exact value when $K \cdot \max |A| \cdot \max |B| < 2^{31}$, for any permutation of the leaves (`integer_parity`); monotonicity of admissibility;
- for memory banks: coprime strides are injective on the lanes (`bank_injective`), and the closed-form padding for 2^m banks is sufficient and minimal;

- for recurrences: the affine monoid is associative with neutral element $(1, 0)$, composition is sequential application, and segmented lifting is associative for any monoid;
- the soundness of the allocation checker (`checkAlloc_sound`);
- the exact envelope decision and its monotonicity in the extent; Higham’s bound on products of $(1 + \delta_i)$;
- the depth of Estrin’s evaluation scheme against Horner’s, computed over the tree.

6.4 Extraction

The extractor (`lake exe vapor-extract`) reads the *elaborated* definitions (the same terms the theorems speak about) and prints Elixir, failing on any construct outside its fragment. The extracted functions are `check_alloc`, `admissible`, `wrap_s32`, `pad_stride`, `bank_of`, `affine_op`, `seg_affine` and `within_envelope`. Lean computes more than 480 conformance vectors that the extracted code must reproduce, and the generated module carries the digest of the Lean sources, which a test checks without needing the Lean toolchain. Extraction happens at build time: production never runs Lean.

6.5 What is tested and what is trusted

Tested but not proved: the encoders (against `binutils` and `spirv-val`, and by execution under QEMU and `lavapipeline`), the RVV interpreter against QEMU at VLEN 128, 256 and 512, bit-for-bit equality across substrates, and fault containment. Trusted: the BEAM and the Linux kernel, the extractor’s printer and its six-line prelude, the IEEE-754 standard model, and the Vulkan driver, contained in its own process.

7 Models: the airlock

7.1 The contract, not the family

A program in the algebra is self-describing: its inputs, outputs and state are typed terms. What the core needs to know about a model is therefore the *shape of its program*, not its family. The model airlock (`Vapor.Lock`) is the only place where a family is known. Inside, there is only a `Vapor.Lock.Spec`, the contract, and a program checked against it.

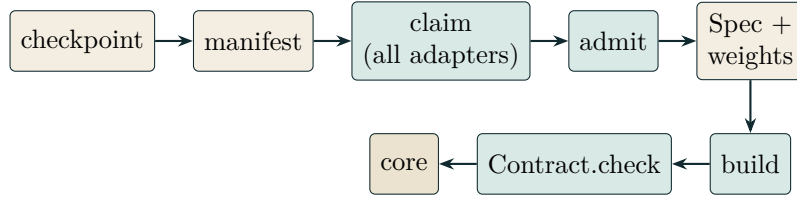


Figure 7.1: Admission. A defective adapter is stopped at the boundary, not inside the engine.

contract	inputs	outputs	state
<code>:causal_lm</code>	tokens and positions (and, by option, sampling, a block table, a slot, soft tokens)	logits and/or hidden rows	paired $\{x, x_{\text{next}}\}$ of equal sorts
<code>:encoder</code>	rows, horizon	hidden rows (pooled, logits)	—
<code>:codec</code>	rows $\leftrightarrow$ codes	same	—
<code>:map</code>	rows	rows	—

Table 7.1: Program contracts. A spec also carries widths, vocabulary, end tokens, modalities, the features its builder accepts, and the digest of the admitted configuration.

Three properties are enforced. **One owner per checkpoint**: every adapter answers a claim with a score, a near miss with its reason, or nothing; with no owner, the refusal lists every near miss and its fix. **The contract at the airlock**: the built program is checked before the core sees it. **No family in the core**: a test reads the atom tables of the compiled core modules and fails if any of them references an adapter or a family’s configuration module.

7.2 Where names live

A checkpoint computes a function, and someone must write that function once in the algebra; that code is mathematics. A checkpoint is also *spelled* a certain way (a `model_type`, configuration keys, tensor names, fused matrices), and a spelling is data about a family.

Adding a family costs, from cheapest to dearest: an **alias** (data only: key renamings, forced and required values, forbidden keys, tensor renamings by template, splits of fused tensors sized by products of configuration fields; an alias can narrow what its base accepts, never widen it); a **blueprint** (a few lines translating configuration into the knobs of an existing topology); a

tier	holds	names it may use	enforced by
core	algebra, compiler, emitters, runtime, engine, deciders	none	atom-table test
topologies	the function, once per <i>kind</i> of network: decoder, delta-rule hybrid, encoder, encoder-decoder, Mamba, Mamba-2, DiT, U-Net, VAE, codec	what they compute	a test that no module name carries a product or family name
spellings	claims, aliases and refusals	product and family names, here only	aliases and refusals are data
agent protocols	the wire format of a hosted model’s API	the vendor’s protocol name	the single allowance

Table 7.2: The naming rule.

topology (a new program built from existing operators, no new kernel). A **refusal** is also data: a family whose mathematics differs from every topology is refused by name, with the reason.

Admission can run on headers alone: `Vapor.Lock.preflight/2` takes a configuration and the safetensors headers of the shards (names, dtypes, shapes, no data) and returns the airlock’s verdict, including missing and unexpected tensors, before a byte of weights is read.

7.3 A worked example: the delta-rule hybrid

The delta-rule hybrid topology (`Vapor.Lock.Adapters.DeltaHybrid`) shows what admitting a frontier architecture involves. Its layers alternate a gated delta-rule attention with a latent attention without positions, combine blocks through attention over earlier block outputs, and route tokens to a latent mixture of experts.

Gated delta rule. Per head, the state $S \in \mathbb{R}^{d_k \times d_v}$ evolves as

$$S_t = (I - \beta_t k_t k_t^\top) \text{Diag}(\alpha_t) S_{t-1} + \beta_t k_t v_t^\top, \quad o_t = S_t^\top q_t,$$

with a per-channel forget gate $\alpha_t = \exp(g_t)$ bounded as $g_t = g_{\min} \sigma(e^A z_t) \in (g_{\min}, 0)$. For prefill in chunks, the recurrence is equivalent to a chunkwise form built on the UT transform: with Γ the cumulative decays inside a chunk and L_s the strictly lower part of the decayed key Gram matrix,

$$T = (I + \text{diag}(\beta) L_s)^{-1}, \quad U = T \text{diag}(\beta) V, \quad W = T \text{diag}(\beta) (K \odot \Gamma).$$

The two forms agree to $3 \cdot 10^{-16}$ in float64; the bounded gate is what keeps the chunk’s reciprocal decay finite in float32 over a 16-token tile, where an unbounded gate overflows.

Latent attention. The latent attention keeps only the compressed latent in the cache (`kv_lora_rank` numbers per token and layer) and absorbs the key projection into the query through a grouped contraction, so decoding never decompresses keys.

Experts. Routing is a sigmoid score plus a bias, top- k by exact counting, shared experts, and a projection into a latent width with a normalisation before the up-projection; the gated activation soft-caps both branches, $\beta_1 \tanh(g/\beta_1) \sigma(g) \cdot \beta_2 \tanh(u/\beta_2)$.

Weights in MXFP4. An MXFP4 value is $\pm\{0, \frac{1}{2}, 1, \frac{3}{2}, 2, 3, 4, 6\} \cdot 2^{s-127}$: a two-bit significand times a power of two. Every such value inside binary32’s range is a binary32 value, so decoding is *exact*, and a model computes the same bits from MXFP4 experts as from any other copy of the same values. Elements that would overflow and NaN scales are refused by name.

Checks. An independent float64 reference, written from the equations alone, computes two small instances its own way (no caches, decompressed latent keys, the whole sequence at every step). The compiled program agrees with it to a relative error of $4.4 \cdot 10^{-6}$ (and $1.35 \cdot 10^{-5}$ with the soft caps active, a low-rank query and MXFP4 experts); greedy decoding is identical; the native worker equals the oracle bit for bit; and the controls (forgetting the delta-rule state or the latent cache at every token) fail, as they must.

Balanced routing, certified. The router’s balanced assignment (each token k experts, each expert $q = mk/n$ tokens) is a bipartite b -matching whose constraint matrix is totally unimodular, so its linear relaxation is integral. **Vapor.Train.Balance** solves it with the exact rational simplex and checks the dual certificate. Coordinate descent on the dual reaches the certified optimum when each threshold is placed between the k -th and $(k+1)$ -th entries; placing it *at* an entry creates ties that leave most fixed points unbalanced.

7.4 The substrate airlock

A substrate receives canonical programs only after it has been measured. The probes of **Vapor.Substrate** produce a verdict (`:canonical`, `:envelope` with a numerical fingerprint, or `:refused`) and the dispatcher respects it. The fingerprint records fused multiply-add, flush-to-zero and denormals-are-zero behaviour, reduction order, mantissa bits, signed zero, NaN handling, division and the transcendental functions; each field that departs from the oracle is named. Metal (through an MSL shim), StableHLO devices through PJRT, cluster nodes and other operating systems enter the same way.

7.5 Renewal and merging

Models are values under Merkle roots. **Palingenesis** renews a model plank by plank: a plank enters through a contract gate, a Fisher–Rao drift brake measured on anchor sequences, a paired test on target sequences and caller invariants; generations are published by read-copy-update on immutable values, and the lineage is a chain of signed records that re-derives from the weights. **Merging** combines checkpoints by declared methods (linear, SLERP, TIES and others), disk to disk, one tensor at a time, with a signed receipt; a diagnostic mode measures the regime before merging, and merging by measurement chooses among methods on held-out text.

8 The network airlock

8.1 The rule

vapor has no network access of its own. The one exception is the agent backends, which speak to the hosted model a person configured. Everything else that comes from outside comes through the **siphon** (`Vapor.Siphon`), under one rule: *an agent never connects on its own; only the person opens the tap.*

8.2 Fetchers

The person declares fetchers in `$VAPOR_HOME/siphons.json`. A fetcher is any program (the official Python client of a model hub, `aws s3 cp`, `rsync` to a private cluster) with an argument template, the environment variables it may see, a deadline and a byte cap:

```
{
  "fetchers": [
    {
      "name": "hub",
      "argv": ["python3", "fetch.py", "{ref}", "{out}"],
      "env": ["HF_TOKEN"],
      "timeout_s": 600,
      "max_bytes": 20000000000,
    },
    {
      "name": "range",
      "argv": ["sh", "-c", "curl -r {range} ...", "{ref}", "{out}"]
    }
  ]
}
```

who	what they can do
the person, at a terminal	<code>vapor siphon run NAME REF; approve ID; headers; preflight</code>
an agent (MCP)	<code>siphon_propose</code> : queue a request with its reason; nothing is fetched
the browser console	nothing: the verb is not offered

Table 8.1: Who opens the tap.

8.3 What a fetch can do, and what comes in

A fetch runs as an operating-system port with only the declared environment variables, a deadline and a byte cap; exceeding either kills it. A reference that could be read as an option is refused, and a reference is always one argument, never a shell word. What lands goes through the format airlocks (safetensors and GGUF headers parsed and bounded, JSON decoded) and may be pinned by SHA-256. Every fetch leaves a receipt with the exact argument vector, the digests and the formats found; nothing lands from a malformed file, a wrong digest or a failing fetcher.

8.4 Headers before data

A remote safetensors file can be admitted from its header alone, with two ranged reads: the first eight bytes give the header length, the next read gives the header. `vapor siphon preflight NAME CONFIG SHARD...` fetches the configuration and every shard’s header and asks the model

airlock whether the checkpoint can run, naming what is missing or unexpected, before a byte of weights is fetched.

8.5 Streaming weights: the arithmetic

Decoding reads every active weight once per token, so the time per token is bounded below by active bytes divided by bandwidth.

model	bytes per token	RAM 50 GB/s	NVMe 3 GB/s	net 20 MB/s
15 GB dense	15 GB	0.3 s	5 s	12.5 min
2.8 T hybrid (MXFP4 experts)	≈ 146 GB	≈ 3 s	≈ 50 s	≈ 2 h

Table 8.2: Streaming weights over a network does not serve conversation. It serves work that reads the weights once for a whole input: one prefill pass to score, classify or embed a long document.

9 Deciders with certificates

9.1 One pattern

Every decider follows the same pattern: *a search proposes, a checker decides*. The search may be heuristic, fast and complicated; the checker is small, separate (it shares no code with the search) and reads only the certificate. Anyone’s proposal (a person’s, a program’s, a language model’s) goes through the same checker, so a model that proposes nonsense simply has its proposal refused.

problem	procedure	certificate, and its check
propositional logic (DIMACS, formulas through Tseitin)	CDCL: two watched literals, first-UIP learning, minimisation, Luby restarts	a model evaluated clause by clause, or a DRUP refutation checked backwards by <code>Vapor.Logic.DRUP</code>
finite combinatorics (Schur, van der Waerden, Ramsey, pigeonhole, queens)	encoding and CDCL, threshold search	a witness below the threshold checked against the definition, and a DRUP refutation at it
equational theories	Knuth–Bendix completion, lexicographic path order	a convergent rewriting system; normal forms with derivations
polynomial geometry	Buchberger with criteria; Rabinowitz’s trick	the basis $\{1\}$ for an implication, or the remainder and the non-degeneracy conditions
linear programs	two-phase simplex over $\mathbb{Q}$	<i>optimal</i> : primal x and dual y with $c^T x = b^T y$; <i>infeasible</i> : a Farkas vector; <i>unbounded</i> : a feasible point and an improving ray
integer programs	branch and bound over the exact simplex	the tree: splits that cover the integer points, a Farkas or dual vector at every leaf, the incumbent
causal identification	the ID algorithm (complete), back-door criterion, d-separation	the estimand, or a hedge checked structurally
polynomial positivity on a box	Bernstein subdivision in exact integers	a replayable subdivision, or the exact point that refutes; barrier certificates by their three conditions
circuit equivalence over $\text{GF}(2)$	truth table up to 16 inputs, otherwise a SAT miter; word-level identities by Gröbner reduction over $\mathbb{Z}$	DRUP-checked proof, or a shrunk counterexample
contracts (deontic norms)	Tseitin and CDCL over the clauses and facts	antinomies with a scenario; consistency proved by DRUP

Table 9.1: The deciders. All arithmetic is exact.

9.2 Selected details

Integer programs. The certificate of an optimal or infeasible integer program is the branch-and-bound tree itself. `MIP.check/2` verifies that the splits partition the integer points of each

node, that every pruned leaf carries either a Farkas vector for its rows (infeasible) or a dual-feasible vector whose bound does not beat the incumbent (by weak duality), and that the incumbent is feasible and integral. On random programs the answers equal brute force, and forged certificates are refused.

Causes. A causal question is asked of a stated diagram, with explicit hidden confounders. Identification returns the estimand, for example $\sum_m P(m | x) \sum_{x'} P(x') P(y | x', m)$, or a hedge that proves non-identifiability; on random structural models, estimands evaluated in exact rationals equal the true intervention, and the naive $P(y | x)$ serves as the control. No verdict says anything without its diagram, and the system says so with every answer.

Circuits and silicon netlists. Rebis decides whether two circuits compute the same function, reads AIGER, BLIF and structural Verilog, and computes algebraic normal forms and word-level identities. Qālib reads netlists of the open sky130 cell library, maps circuits to cells, and proves every mapped netlist equal to its specification before it is printed; it finds, by SAT, a 20-bit trigger in a 21-input adder that 4,096 random patterns miss. It is translation validation of the open flow, not a chip compiler.

Silent corruption and order. Two numerical tools guard computations at scale. The cupel checks matrix products by a randomised projection identity in $O(n + k)$ and reports a per-bit detection profile, catching a flipped bit in a product. The amalgam sums numbers exactly and rounds once, so the result does not depend on order, topology or the number of workers; naive sums in several orders are shown beside it.

9.3 Finance

The finance desk applies the same rule where the pain is verifiability. Money is exact decimal; business calendars follow rules that match QuantLib day by day from 1990 to 2078; curves come with a repricing certificate; option surfaces are checked for static arbitrage by the exact simplex, with a Farkas vector as proof of absence; backtests pass four noise gates (including combinatorially symmetric cross-validation); an order book is replayed against a naive judge written separately. The Monte Carlo engine compiles its trajectory step (generator included) to the algebra: the generator is chosen to fit exactly in binary32 (Lehmer products below 2^{23} , the modulo by a rounding trick), so the paths are the same bits on every substrate and at every thread count, checked against the oracle in the answer itself.

10 Languages

10.1 Almizan: claims that do not compile until they are decided

Almizan is a small language in which each claim says *what kind* it is (its root) and *how it may be used* (its pattern, *wazn*). A claim marked as proved does not compile until a decision procedure proves it, or refutes it with the point that breaks it.

One tree, two scripts. A program is a tree; Latin and Arabic are bijective printings of it, with the same canonical hash: the projection changes keywords and digits, never names. Below, a law written with Latin names in the Latin printing, and the same law as an Arabic-speaking author writes it, with Arabic names, in the Arabic printing.

```
(claim energy (root H-f-Z) (wazn burhan)
  (inputs (x q) (v q))
  (field (x v) (v (- x)))
  (proof conserved)
  (body (+ (kinetic v) (* 1/2 x x))))
```

(دعوى طاقة (جذر ح-ف-ظ) (وزن برهان)
 (مدخلات (س نسبي) (ع نسبي))
 (حقل (س ع) (ع (- س))
 (برهان محفوظ)
 (تنفيذ (جمع (حركية ع) (* ١/٢ س س))))

Keywords have both forms; numbers use Western digits in one script and Arabic-Indic digits in the other; names are Latin or Arabic but never mixed; roots in Latin use the Buckwalter transliteration, which is lossless. `read(print(t)) = t` holds in both scripts.

root	domain	what can be claimed
H-s-b	calculation	a value; identities; bounds and positivity on a box
H-f-Z	conservation	a quantity constant along a field $\dot{x} = f(x)$
n-q-l	transition	a system of Boolean steps and its invariant
k-t-b	record	a stored value, named by its hash
s-b-b	cause	an interventional query on a stated causal diagram

Table 10.1: Roots. The patterns are *fāʿil* (a pure function, lowered to the compiler), *mafcūl* (a stored value) and *burhān* (proved: nothing runs until the obligation is met). Meaningless pairs are refused.

A morphological type system.

proof	decider	certificate
(identity E)	polynomial normal form over $\mathbb{Q}$	both sides normalise to one polynomial
conserved	$\dot{H} = \nabla H \cdot f$ normalised over $\mathbb{Q}$	the zero polynomial, or the remainder and a point where it is not zero
nonneg, pos, (bounded a b)	Bernstein subdivision in exact integers	a replayable subdivision, or the refuting point
invariant	SAT by induction	a DRUP proof checked separately, or the counterexample
identifiable, (adjustment z), separated	ID, back-door, d-separation	the estimand, or a hedge, or the open path

Table 10.2: Almizan’s deciders.

Proof by decision. The damped oscillator shows a refutation: with the field $\dot{v} = -x - \frac{1}{10}v$, the decider reports $\dot{H} = -\frac{1}{10}v^2$ and the point $v = 1, x = 0$ where it is not zero. A canonical formatter keeps comments as glosses attached to the declaration they precede; it does not reorder declarations, because the order is the author’s argument and the hash already ignores layout and script. Almizan exports its claims to core Lean (**Rat, grind**), and a test tier checks that Lean proves what **vapor** proved and rejects what it refuted.

10.2 Alembic: open input, sanitised

Alembic is the language in which a person or a model writes a problem for the search engine: a space and an objective, a claim to refute or prove by enumeration, a game. The input is free, so the limits belong to the machine:

risk	defence
infinite loops, deep recursion	fuel charged on every call and iteration; depth bounded
huge numbers and lists	integers up to 65,536 bits; lists of two million elements; bounded ranges
memory and time	the hermetic seal (below)
the atom table	identifiers stay binaries, never atoms
code in place of data	a literal reader that accepts only data and constant arithmetic

Table 10.3: Alembic’s defences. There is no I/O, no clock and no randomness beyond a deterministic hash: a program is a pure function of its text.

The evaluator compiles the tree once into nested closures, so a verifier called a hundred thousand times during a search pays for the tree walk once.

10.3 Athanor and the Touchstone

Athanor is a general search engine for problems written in Alembic. It runs a portfolio of strategies under a bandit that allocates the budget by measured progress, beside a random-search control with the same budget, and records every evaluation in a journal whose Merkle root identifies the run; the same seed gives the same root. A result is a certificate (a counterexample, an exhaustive enumeration, an incumbent) that the **Touchstone** re-checks without trusting the search: a forged value is refused. Games written in Alembic are solved exactly by negamax when they fit, or searched by Monte Carlo tree search. Problems whose objective is measured outside (a training run, a simulation) are optimised by asking the person to measure the proposed candidates.

10.4 The hermetic seal

One containment primitive, `Vapor.Hermetic.seal/2`, runs every job on untrusted input: an Alembic evaluation, a document ingestion, a console terminal session, an Athanor run. The job runs in its own process with a heap cap that *includes* off-heap binaries (which on the BEAM live outside the process heap and would otherwise escape a cap) and a deadline. A memory bomb is killed and the caller survives. The seal bounds memory and time; it does not defend against code that already has access to the BEAM itself.

11 The evidence discipline

11.1 A value, a control, a threshold

Every quality check pairs three things: a **value** computed by the system, a **control** that a broken, naive or lucky implementation would produce, and a **threshold** that separates them. The question each check answers is the one that matters for open input: *would the same answer have come out if there were nothing to find?*

check	value	control (must fail, or be caught)
render, white furnace	a sphere of albedo a in a uniform environment shows exactly a	–
render, gradient furnace	$a(\frac{1}{2} + n_y/3)$ under a sky that brightens upward	the biased estimator gives $a(\frac{1}{2} + n_y/4)$ and is caught
ink style	a hard shadow at exactly albedo $\times$ ambient	the same scene without the occluder: the pixel is lit
ODE integration	Dormand–Prince to 10^{-8}	fixed-step RK4 at $h = 0.1$: error above 10^{-5}
PDE	Crank–Nicolson observed order 2	backward Euler: order 1
SAT	a witness and a DRUP refutation	a truncated proof: rejected
scaling law	the planted exponent inside the confidence interval	shuffled losses: holdout error above 1 %
calibration	a calibrated model not flagged	an over-confident one flagged
recommendations	a significant, minimum-size gain over the biases	shuffled ratings: at the mean

Table 11.1: A few of the quality checks; the suite holds about one hundred and fifty.

Noise gates (for text, images and audio) are fitted to negative controls (noise, shuffled signal, repetition) and positive ones (held-out real signal), and *refuse to exist* if the two overlap.

11.2 The Assay: statistics for AI research

The Assay is a set of tools whose answers say whether a difference is signal or noise: paired comparisons with bootstrap intervals and sign-flip tests; leaderboards with rank intervals; calibration against its own sampling floor; inter-annotator agreement (Krippendorff’s α); the position bias of a model used as a judge; train/test contamination by n -gram overlap; near-duplicate detection by MinHash verified by exact Jaccard; scaling-law fits with confidence intervals; which layer of an encoder a linear probe should read (chosen on validation, read once on test, with a shuffled-label control); and detection scored by cgF1, where boxes are matched by an optimal assignment solved exactly by the rational simplex on a totally unimodular program, so the matching is certified optimal.

11.3 The assurance ledger

Claims about the system are kept in a ledger (`Vapor.Assurance`) with a status (*proved*, *checked*, *tested*, *owed*), the evidence (files that must exist) and the limits. The human-readable document is generated from the ledger, and a test fails if cited evidence disappears or if the document drifts from the ledger. Documentation that names a module, a function or a repository path that does not exist fails another test.

11.4 Tests

The suite holds about a thousand tests in 159 files. Its structure follows the evidence it gathers.

tier	what it adds
unit and property	the BEAM alone; random programs, round trips, invariants
native	the worker on the host: every kernel against the oracle, bit for bit
cross	QEMU for AArch64 and RISC-V; the RVV interpreter against QEMU
GPU	lavapipe (Vulkan on the CPU); headless Chromium for the console and the WebGL tracer
differential	independent references: <code>transformers</code> , <code>diffusers</code> , NumPy, Tesseract, QuantLib, ngspice, python-chess and others, each a tier that runs when its tool is present
formal	Lean builds, the extraction is byte-identical, exported Almizan claims agree

Table 11.2: Test tiers. A tier whose tooling is missing is excluded and reported, never silently passed.

For the edit loop, `mix vapor.test` keeps a content-addressed cache. A test file’s key is the SHA-256 of the file, the support files and fixtures, `priv/`, the toolchain versions, the excluded tiers, and the compiled bytecode of every module the file can reach, transitively through the modules’ atom tables. A file is recorded only when every test in it passed; changing one module reruns only the files that can reach it. The cache is for the edit loop, not for releases: `mix test` ignores it.

12 Interfaces

12.1 One interpreter

Every capability is a verb of one interpreter, the *Dīwān*, which serves the command line (*bin/vapor*), the terminal user interface and the browser console's terminal, so the three answer alike. Verbs read files or standard input, write text on a terminal and JSON in a pipe, and return exit statuses with meaning: 0 done and positive, 1 done and negative (refuted, not found, verification failed), 2 usage, 3 invalid input, 4 failure. In the console, the interpreter runs jailed: files come from the session, and verbs that would touch the server's disk or network (renewing a model from disk, writing a rendered file, the network airlock, the editor) are not offered.

```
vapor wzn check energy.wzn          # Almizan: decide every claim
vapor athanor run golomb.nbq | vapor verify golomb.nbq -
vapor logic knapsack.lp | jq .objective
vapor qalib check adder.net synth.v  # exit 1 with the input that tells them apart
vapor render studio.txt --ink --out sketch.png
vapor siphon preflight hub config.json model-00001-of-00002.safetensors ...
```

12.2 The console

A single offline page served by the BEAM: no external font, script or request. It groups its panels by what one does (ask, solve, simulate, make, read, measure, trust) and alphabetises them in the language shown, English or Portuguese. The workspace detects the kind of problem typed and shows the search live in the *furnace* (sparks per evaluation, the best-so-far line, the random control, the portfolio's allocation) and the verdict on the *touchstone*. The render desk traces light progressively on the viewer's GPU (WebGL), keeps the scene text as the source of truth (dragging the image rewrites the camera line), and compares its mean radiance with the server's reference; the same scene can be drawn in ink.

12.3 Agents

MCP. `mix vapor.mcp` exposes 27 tools: the language and the search (`alembic_eval`, `athanor_run`, `athanor_verify`), the deciders (`logic_check`, `rebis_check`, `aludel_decide`, `tabula_analyze`, `qalib_check`), science and statistics (`crucible_run`, `assay_run`), the workbench and engineering desks, finance, boards, rendering, recommendations, the studio, and `siphon_propose`, which only queues a request for the person. An agent proposes; the checker decides.

Agents as proof objects. An agent's definition (`Vapor.Agent.Spec`) is a value with a digest and a lineage; its tools carry an effect class (`pure` re-executed on replay, `observe` read from the record, `act` never re-executed and run only with a grant fixed in the digest). A run is a hash-chained journal with a Merkle root, attested by the executing node's Ed25519 key; replaying a run with a local model recomputes every decision on any substrate and must end at the same head. The store writes ahead of every step and announces every action before it

happens, so an action happens at most once with an idempotent recipient. Personal data are sealed per data subject, and destroying the key erases them everywhere while the chain still verifies.

12.4 Editors

The language server. `vapor lsp` speaks LSP over standard I/O to any editor: for Almizan, diagnostics with the refuting point, hover with the decider, completion in the file's script, symbols and definitions, canonical formatting that keeps comments, the other script as a command, and inlay hints that put each claim's verdict at the end of its line.

Al-Qalam. `vapor qalam FILE` is a small editor of the system's own, in the terminal: a subset of vi; each claim's verdict in the gutter ($\checkmark$, $\times$, $?$, $!$), rechecked after every change made outside insert mode; numbers that `Ctrl-A` and `Ctrl-X` step (a fraction by its own unit), so a coefficient can be walked until a law starts or stops holding; bracket and top-level motions; canonical formatting and the two scripts; and an undo tree whose states are identified by the SHA-256 of their parent and their text, so branches are kept and identical histories have identical identities. Its core is pure (state and keys to state, state and size to screen) and is tested without a terminal and through a real pseudo-terminal.

13 Limits

What the system does not do, or has not been shown to do, is part of its specification.

Hardware. RVV and NEON code is validated under QEMU, and the RVV interpreter against QEMU; Vulkan is validated under lavapipe. No physical RISC-V vector hardware, discrete GPU or Apple Silicon machine has run the code. The cooperative-matrix path of the SPIR-V backend is emitted and validated but has not executed. Scaling to many cores and hardware counters are measured where a PMU exists.

Numerics. Under the canonical policy $+$, $-$, $\times$ and $\div$ are IEEE operations; the transcendental functions are identical on every substrate but within one or two ulps, not correctly rounded. The oracle and the envelope are exact and therefore slow, which limits the size of the probes in the differential rungs (the bounds extend to the maximal extents by monotonicity). Normalised text depends on OTP's Unicode version, which is recorded in digests rather than eliminated.

Performance. The 4-bit matrix-vector product is bound by instruction issue. The Vulkan daemon runs self-contained units without resident sessions, and its attention recomputes scores in passes without shared memory: correct and bit for bit, not tuned for a real GPU.

Models. Families whose mathematics differs from every topology are refused by name (for example dynamic NTK and LongRoPE rotary scaling). Frontier checkpoints of trillions of parameters are admitted on small instances against independent references; their own released spellings must be read through header-only admission before they are claimed. Hosted models are observations, not functions: their decisions can be recorded and protected, not reproduced.

Certificates. A certificate says what was computed, not whether a model is good. A causal verdict says nothing without its diagram. Qālib proves combinational equivalence of netlists; it does not make chips. The hermetic seal bounds memory and time, not code with access to the BEAM.

The network airlock. It bounds what a fetch can do and what it brings in; it does not audit the fetchers, which are the person's programs.

Editors. Al-Qalam counts a wide character as one column and leaves right-to-left layout to the terminal; it redraws the whole screen per key, which suits source files, not very large ones.

A Module map

component	modules
algebra	Vapor.Algebra.Term, Vapor.Program, Vapor.Tensor, Vapor.F32, Vapor.Canon, Vapor.CR
oracle and envelope	Vapor.Runtime.Oracle, Vapor.Verify.Envelope
compiler	Vapor.Compile.Rewrite, Vapor.Compile.Lower, Vapor.KIR.{Kernels, Liveness, RegAlloc}, Vapor.Extracted
encoders	Vapor.Emit.{X86, X86.AVX512, ARM, RVV, SpirV, MSL}, Vapor.Export.StableHLO
runtime	Vapor.Runtime.{Worker, Session, Dispatch, Shm, Substrates}, Vapor.Substrate
certificates	Vapor.Certificate, Vapor.Bundle, Vapor.Canonical (CBOR, RFC 8949 §4.2), Vapor.Merkle
engine and serving	Vapor.Engine, Vapor.Recurrent, Vapor.Speculative, Vapor.Sampler, Vapor.Grammar, Vapor.Serve, Vapor.Plug (integrations/vapor_plug)
model airlock	Vapor.Lock, Vapor.Lock.{Spec, Contract, Alias, Adapter}, Vapor.Lock.Adapters.*, Vapor.Quant.MXFP4, Vapor.Ingest.Safetensors, Vapor.Model.GGUF
network airlock	Vapor.Siphon
renewal and merging	Vapor.Palingenesis, Vapor.Merge, Vapor.Train.Balance
deciders	Vapor.Logic (DRUP, LP, MIP, Causal, Knuth–Bendix, Gröbner), Vapor.Aludel, Vapor.Rebis, Vapor.Qalib, Vapor.Tabula, Vapor.Cupel, Vapor.Amalgam
languages	Vapor.Almizan, Vapor.Almizan.{Syntax, Format}, Vapor.Alembic, Vapor.Athanor, Vapor.Athanor.Touchstone, Vapor.Hermetic
laboratory	Vapor.Crucible, Vapor.Assay, Vapor.Solve, Vapor.Engineering, Vapor.Finance, Vapor.Play, Vapor.Bio, Vapor.Render, Vapor.Sketch, Vapor.Studio, Vapor.Docs
evidence	Vapor.Quality, Vapor.Assurance, Vapor.TestCache
agents	Vapor.Agent, Vapor.Agent.{Spec, Journal, Keys, Store, Backend.*}, Vapor.MCP.Server, Vapor.Mind
interfaces	Vapor.Main (and the Dīwān), Vapor.Console, Vapor.TUI, Vapor.LSP, Vapor.Qalam
native	native/src: the worker, the thread pool, the RVV interpreter, the sandbox, the Vulkan and Metal daemons
proofs	proofs/Vapor: integer parity, banks, the affine monoid, the allocation checker, the envelope, Higham’s bound; the extractor